

Calibrated Inference for the Conditional Average Treatment Effect in the Few-Placebo Regime via Gaussian Processes

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Abstract

Estimating how much an intervention helps a given individual — the conditional average treatment effect (CATE) — is increasingly central to decision-making in medicine, economics, and policy, where an estimate is most useful when accompanied by a calibrated uncertainty interval. We study the *few-placebo regime*, in which one treatment arm is much smaller than the other, as arises in unequal-allocation trials and small-holdout A/B tests. The standard estimator in this setting is the X-Learner, and a natural way to obtain credible intervals is to make its second stage Bayesian. We show that these intervals under-cover: they contain the true effect less often than their nominal level. We trace this to a structural cause — the X-Learner’s regression target inherits the bias of a nuisance model fitted to the small arm, so the posterior is centered away from the true effect — and we find that the standard remedy, regressing an orthogonal doubly-robust score, is also unreliable here, since the regime’s limited overlap leaves the estimator either highly variable or, once stabilized, biased once more. Both consequences reflect a pattern that extends beyond causal inference: a separately estimated variance is attached to a point estimate of a hard-to-learn quantity, and the point estimate’s bias is not captured by that variance. We propose GP-CATE, which models each arm’s outcome surface with a Gaussian process, so the scarce arm’s uncertainty enters the posterior directly rather than as an unmodelled bias. Across synthetic and semi-synthetic benchmarks, GP-CATE attains calibrated coverage where the estimators we compare against — including Causal Forest and BART — do not, at the cost of intervals that are appropriately wide when the data are uninformative.

1 Introduction

Estimating heterogeneous treatment effects — the conditional average treatment effect (CATE) $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$ — is central to data-driven decisions in medicine, economics, and policy. A point estimate of an individual effect, with no calibrated interval around it, cannot support a high-stakes decision: as flexible machine-learning models replace parametric ones, *calibrated uncertainty* has become the binding constraint on deployment.

We focus on the *few-placebo regime*: one treatment arm is much smaller than the other, $N_0 \ll N_1$. This is not a corner case. Unequal-allocation clinical trials assign more patients to the active arm by design; budget-constrained A/B tests hold out only a small control group; rare-regime studies (a stress period, an infrequent exposure) have few units in the rare arm. The canonical estimator here is the *X-Learner* (Künzel et al., 2019), a pseudo-outcome meta-learner designed for arm imbalance.

The X-Learner, however, provides only a point estimate. To attach a credible interval one would naturally make its second stage Bayesian — regress the pseudo-outcome with a probabilistic model and read off the posterior. We show this under-covers. The X-Learner’s treated-arm pseudo-outcome $D_1 = Y - \hat{\mu}_0(X)$ decomposes as $\tau(X) + (\mu_0 - \hat{\mu}_0)(X) + \text{noise}$, and

the imputation error $\mu_0 - \hat{\mu}_0$ is not mean-zero: every regularized machine-learning nuisance trades bias for variance, and because the X-Learner pseudo-outcome is *not Neyman-orthogonal* it inherits that regularization bias to first order. A Bayesian regression on D_1 is centered on a biased estimate of τ , and no variance model moves a misplaced interval. The effect is worst where the X-Learner is meant to be used: in the few-placebo regime $\hat{\mu}_0$ is fit on a small sample. On our synthetic design a nominal-95 % interval covers 0.88 at $N_0 = 30$.

The standard remedy for first-order nuisance bias is to regress an *orthogonal* pseudo-outcome — canonically Kennedy’s (Kennedy, 2023) doubly-robust score. This too fails in the few-placebo regime. The regime forces extreme propensities (a tiny control arm means $1 - \pi$ is small), and the orthogonal score’s “second-order” remainder is amplified by a $1/(1 - \pi)^2$ factor that grows large precisely here; moreover, even with the propensity *known exactly*, an efficient inverse-variance-weighted second stage downweights the high-variance control rows to near-zero — and those are the rows whose augmentation cancels the nuisance bias. The orthogonal-score estimate’s bias is, in our experiments, indistinguishable from the non-orthogonal one’s. Orthogonalization fails because the few-placebo regime itself induces the poor overlap on which it depends.

These two failures look unrelated, but they share a single root cause: each attaches a variance to a *point estimate* of the hard placebo nuisance μ_0 — a regularized tree fit on the few control units — and a point estimate’s bias is invisible to any variance computed after the fact. The same defect afflicts any repair that keeps a point estimate at its core, including modelling the regularization bias and subtracting it. The defect is structural, and it points to its own fix.

Our proposed estimator, GP-CATE, addresses this defect at its source. Instead of a point estimate plus a post-hoc variance, it models each arm’s outcome surface with a *Gaussian process*. The GP posterior over the function μ_w is calibrated by construction: wide where that arm’s units are sparse, narrow where they are dense. The placebo arm’s small sample therefore enters the posterior directly as uncertainty rather than as an unmodelled bias, and the CATE posterior is simply the difference of the two arm posteriors, $\tau(x) \sim \mathcal{N}(m_1(x) - m_0(x), s_1(x)^2 + s_0(x)^2)$. The estimator is not itself new — a Bayesian T-learner with GP arms is close to Alaa and van der Schaar (2017); our contribution is to identify *why* the meta-learner approaches mis-calibrate in the few-placebo regime, and to show that a fully-Bayesian nuisance is what restores calibrated coverage.

Empirically, on a unified protocol — a linear and a non-linear synthetic design, and the IHDP semi-synthetic benchmark sub-sampled to mirror $N_0 \ll N_1$ — GP-CATE is the only estimator we test that is calibrated throughout (0.94–0.99 coverage). The two most common CATE estimators that provide intervals — Causal Forest and BART — both under-cover in this regime, to as low as 0.34 on IHDP. The cost of GP-CATE’s calibration is interval width: on the 25-dimensional IHDP covariates, thirty controls do not determine the placebo surface, so the calibrated interval is correspondingly wide — where the alternatives report narrower intervals that do not cover.

In summary, we make four contributions:

- a diagnosis: credible-interval under-coverage for a Bayesian X-Learner in the few-placebo regime is caused by first-order nuisance bias from non-orthogonality (§3);
- a negative result: the doubly-robust / orthogonal-score remedy fails in this regime — through overlap amplification and a conflict between efficiency weighting and unbiasedness — demonstrated empirically, including with the propensity known (§4);
- a root-cause analysis and a method: both failures attach a variance to a point-estimate nuisance; GP-CATE instead models the nuisance with a Bayesian posterior (§5);
- an empirical comparison: on a unified protocol over three designs — including the IHDP benchmark — GP-CATE is calibrated where the alternatives are not, and we report its associated interval width (§6).

The remainder of the paper is organized as follows. §2 reviews the X-Learner, cross-fitting, and orthogonality. §3 develops the diagnosis. §4 establishes the negative result. §5 presents GP-CATE. §6 reports the experiments. §7–§8 discuss and conclude.

2 Background and Setup

We observe i.i.d. tuples $(X_i, W_i, Y_i)_{i=1}^N$ with covariates $X_i \in \mathcal{X} \subseteq \mathbb{R}^d$, binary treatment $W_i \in \{0, 1\}$, and outcome $Y_i \in \mathbb{R}$. Write $N_0 = \sum_i \mathbb{1}\{W_i = 0\}$ and $N_1 = \sum_i \mathbb{1}\{W_i = 1\}$; the few-placebo regime is $N_0 \ll N_1$. Let $\pi(x) = \mathbb{P}(W = 1 \mid X = x)$ be the propensity score and $\mu_w(x) = \mathbb{E}[Y \mid W = w, X = x]$ the per-arm conditional mean. Under unconfoundedness $W \perp\!\!\!\perp (Y(0), Y(1)) \mid X$ and overlap $0 < \pi(x) < 1$ a.s. (Rosenbaum and Rubin, 1983), the CATE $\tau(x) = \mu_1(x) - \mu_0(x)$ is identified.

The canonical meta-learner for this setting is the X-Learner of Künzel et al. (2019), which estimates τ in three steps. (i) Fit per-arm conditional means $\hat{\mu}_0, \hat{\mu}_1$ by cross-fitted machine learners. (ii) Form cross-arm pseudo-outcomes,

$$D_1^{(i)} = Y_i - \hat{\mu}_0(X_i) \quad (W_i = 1), \quad (1)$$

$$D_0^{(i)} = \hat{\mu}_1(X_i) - Y_i \quad (W_i = 0). \quad (2)$$

(iii) Regress D_w on X within each arm and combine by propensity weighting. The X-Learner is designed for arm imbalance: cross-arm imputation transfers structure from the large arm to the small one, and it out-performs the S- and T-Learner when the arms differ greatly in size. To avoid in-sample bias from plugging $\hat{\mu}_w$ into (1)–(2), we use K -fold cross-fitting (Chernozhukov et al., 2018) throughout: nuisances are trained on $K - 1$ folds and evaluated on the held-out fold, so each unit’s pseudo-outcome uses a nuisance not fit on that unit.

A central concept for what follows is *Neyman-orthogonality*. A pseudo-outcome is Neyman-orthogonal in its nuisances when the conditional mean of the pseudo-outcome is first-order insensitive to nuisance error — its Gateaux derivative in every nuisance direction vanishes, so only second-order products of nuisance errors survive (Chernozhukov et al., 2018). The X-Learner pseudo-outcomes (1)–(2) are *not* orthogonal. The canonical orthogonal alternative is the doubly-robust (DR) score (Kennedy, 2023; Robins et al., 1995),

$$\begin{aligned} D_i^{\text{DR}} = & \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) \\ & + \frac{W_i(Y_i - \hat{\mu}_1(X_i))}{\hat{\pi}(X_i)} \\ & - \frac{(1 - W_i)(Y_i - \hat{\mu}_0(X_i))}{1 - \hat{\pi}(X_i)}, \end{aligned} \quad (3)$$

whose conditional mean equals $\tau(X)$ up to a remainder that is a *product* of nuisance errors. Regressing D^{DR} gives the DR-Learner (Kennedy, 2023).

Both pieces of (3) divide by $\hat{\pi}$ or $1 - \hat{\pi}$, so the behaviour of the DR score is governed by *overlap*: how far $\pi(x)$ stays from 0 and 1. The few-placebo regime is, by construction, a poor-overlap regime — a small control arm means the effective probability of being a control is small, so $1 - \pi$ is small. §4 shows that this is what causes the orthogonal score to fail.

Finally, we will use Gaussian-process regression. A Gaussian process (GP) places a prior over functions; conditioned on data it returns a full posterior — a mean and a pointwise variance — over the regression function, not merely a point estimate (Rasmussen and Williams, 2006). Its posterior variance is large where data is sparse and small where data is dense, and under standard conditions GP posteriors contract at minimax rates and their credible sets attain frequentist coverage at the nominal level (Ghosal and van der Vaart, 2007). GPs have been used for treatment effects before, notably the multi-task GP model of Alaa and van der Schaar (2017).

§5 uses a GP to model each arm’s outcome surface, so that the small placebo arm’s uncertainty is represented directly in the posterior rather than estimated as a point and corrected after the fact.

Throughout, we assume the regularity conditions of Chernozhukov et al. (2018) for the nuisance estimators and the identification conditions above. The diagnosis and the negative result (§3–§4) concern a second-stage CATE regression in a finite-dimensional basis $\phi : \mathcal{X} \rightarrow \mathbb{R}^p$, $\tau(x) = \phi(x)^\top \beta$; the proposed method (§5) replaces that basis with the GP above.

3 First-Order Bias of the X-Learner Pseudo-Outcome

To attach credible intervals to the X-Learner we make its second stage Bayesian — regress the pseudo-outcome with a probabilistic model and report the posterior over $\tau(x)$. This section shows the resulting intervals under-cover in the few-placebo regime, and isolates the cause: the pseudo-outcome’s conditional mean is not τ , so the posterior is centered away from τ .

For a treated unit i , substitute $Y_i = \mu_1(X_i) + \epsilon_i$ with $\mathbb{E}[\epsilon_i | X_i] = 0$ into $D_1^{(i)} = Y_i - \hat{\mu}_0(X_i)$:

$$D_1^{(i)} = \underbrace{\mu_1(X_i) - \mu_0(X_i)}_{=\tau(X_i)} + \epsilon_i - \underbrace{(\hat{\mu}_0(X_i) - \mu_0(X_i))}_{=: \delta_0(X_i)}. \quad (4)$$

The X-Learner’s second stage treats D_1 as an unbiased draw of $\tau(X_i)$. It is not. The decomposition is often read as “unbiased under a correctly specified nuisance”: if $\hat{\mu}_0$ were conditionally unbiased, $\mathbb{E}[\delta_0 | X] = 0$. But machine-learning nuisances are *not* conditionally unbiased — every regularized learner (boosting, forests, penalised regression) trades bias for variance, so $\mathbb{E}[\hat{\mu}_0(x)] \neq \mu_0(x)$. Writing the regularization bias $b_0(x) = \mathbb{E}[\hat{\mu}_0(x)] - \mu_0(x)$,

$$\mathbb{E}[D_1^{(i)} | X_i] = \tau(X_i) - b_0(X_i). \quad (5)$$

The dependence on the nuisance is *first-order*: a perturbation $\hat{\mu}_0 \mapsto \hat{\mu}_0 + h$ shifts $\mathbb{E}[D_1]$ by $-h$, one for one. In semiparametric language the map $\hat{\mu}_0 \mapsto D_1$ is not Neyman-orthogonal: its Gateaux derivative does not vanish. The bias b_0 is largest exactly in the few-placebo regime, where $\hat{\mu}_0$ is fit on a small sample and heavily smoothed.

The bias displaces the credible interval rather than merely widening it. A Bayesian regression on D_1 produces a posterior for τ centered on the projection of $\mathbb{E}[D_1 | X] = \tau - b_0$ onto the basis. The credible interval is therefore displaced by b_0 . This is not a defect a variance model can repair: a noise model sets the posterior *width*, while the bias sets its *location*, and these are separate factors of the Gaussian posterior. Even an exact noise covariance leaves the interval centered on $\tau - b_0$. The standard correction of inflating the interval (a bootstrap variance times a constant) raises coverage only by widening it until it encompasses a known bias, which is not principled inference.

The experiments in §6 measure this. On the linear synthetic design the X-Learner pseudo-outcome carries a systematic bias of 0.15 at $N_0 = 30$ — about 55 % of the posterior spread — and a nominal-95 % credible interval covers 0.88 (Table 1). The bias decays to 0.03 at $N_0 = 500$ as the placebo nuisance sharpens, and coverage returns to near-nominal. The miscalibration is genuine and concentrated in precisely the few-placebo regime for which the X-Learner is intended.

The standard remedy for first-order nuisance bias is to regress a *Neyman-orthogonal* pseudo-outcome instead of D_1 — canonically the doubly-robust score D^{DR} of (3), whose conditional mean equals τ up to a second-order product of nuisance errors. If that remainder were negligible, a Bayesian regression on D^{DR} would be centered on τ and the coverage problem would be solved. The next section asks whether it is — and shows that, in the few-placebo regime, it is not.

4 Limitations of Orthogonal Scores in the Few-Placebo Regime

The doubly-robust score D^{DR} of (3) is the standard remedy for the first-order bias of §3. This section shows that it does not succeed in the few-placebo regime. We first identify the mechanism — the score’s orthogonality remainder is amplified by poor overlap — and then follow its consequences in two directions. The canonical, unweighted DR-Learner remains unbiased but its variance becomes very large; the natural remedy, inverse-variance weighting, reintroduces the bias. We close by delimiting the scope of the claim.

Conditioning on the cross-fitted nuisances, the DR score’s mean is $\mathbb{E}[D^{\text{DR}} \mid X] = \tau(X) + R(X)$ with

$$R(X) = \left(\frac{\pi}{\hat{\pi}} - 1\right)(\mu_1 - \hat{\mu}_1) - \left(\frac{1-\pi}{1-\hat{\pi}} - 1\right)(\mu_0 - \hat{\mu}_0). \quad (6)$$

This is the “second-order” remainder: each term is a product of a propensity error and an outcome-nuisance error. The orthogonality guarantee is that R is negligible against the $\mathcal{O}(N^{-1/2})$ sampling spread. But that guarantee has a hidden dependence on overlap. A first-order expansion gives $\frac{1-\pi}{1-\hat{\pi}} - 1 \approx (\hat{\pi} - \pi)/(1 - \pi)$, so the control-arm term of R is

$$\left(\frac{1-\pi}{1-\hat{\pi}} - 1\right)(\mu_0 - \hat{\mu}_0) \approx \frac{(\hat{\pi} - \pi)(\mu_0 - \hat{\mu}_0)}{1 - \pi}. \quad (7)$$

The factor $1/(1 - \pi)$ — and, in the contribution to squared error, $1/(1 - \pi)^2$ — diverges precisely in the few-placebo regime, where a small control arm forces $1 - \pi$ small. “Second-order” is an asymptotic statement in N at fixed overlap; here overlap degrades *with* the regime. On our design $1 - \pi \approx 0.06$ at $N_0=30$ — an amplification of order $1/(1 - \pi)^2 \approx 300$. This factor drives the two consequences that follow.

Consider first the canonical, unweighted DR-Learner of Kennedy (2023), which regresses D^{DR} on X by ordinary, *unweighted* least squares. Unweighted regression treats the two arms in the proportions the score itself prescribes, preserving the cross-arm cancellation that makes the score orthogonal; granting that the remainder (6) is controlled (it vanishes when π is known), the DR-Learner is unbiased. That is its theoretical appeal, which we do not contest. In the few-placebo regime, however, this unbiasedness comes at the cost of variance. A control row of D^{DR} carries the inverse-propensity factor $1/(1 - \pi)$, so its conditional variance is inflated by $1/(1 - \pi)^2 \approx 300$ on our design. Unweighted least squares inherits that variance in full. The DR-Learner’s point estimate is then unbiased but very noisy: in our experiments its $\sqrt{\varepsilon_{\text{PEHE}}}$ is 0.63 at $N_0=30$, the largest of any method we test (Table 5). The canonical orthogonal estimator is theoretically unbiased and, in this regime, of little practical use.

Confronted with this large variance, a practitioner may turn to the second consequence: the standard remedy for heteroskedasticity, inverse-variance (GLS) weighting, which down-weights the high-variance rows. The strategy is, however, unsuccessful, and this is so *even when the propensity is known exactly*, in which case $R(X) \equiv 0$ and $\mathbb{E}[D^{\text{DR}} \mid X] = \tau(X)$ holds exactly.

To see why, note that this identity marginalises over W , while the arm-conditional means do *not* equal τ :

$$\begin{aligned} \mathbb{E}[D^{\text{DR}} \mid X, W=1] &= \tau(X) + (\mu_0 - \hat{\mu}_0), \\ \mathbb{E}[D^{\text{DR}} \mid X, W=0] &= \tau(X) - \frac{\mu_0 - \hat{\mu}_0}{1 - \pi}, \end{aligned} \quad (8)$$

and the unbiased combination $\pi \cdot (\cdot \mid W=1) + (1 - \pi) \cdot (\cdot \mid W=0) = \tau$ relies on the two arms entering with their natural proportions: the $(\mu_0 - \hat{\mu}_0)$ bias in the treated rows is cancelled by the control rows. This is the cross-arm cancellation that unweighted least squares preserves. Inverse-variance weighting disrupts it. The control rows, whose variance is inflated by $1/(1 - \pi)^2$, are down-weighted to near-zero — and they are exactly the rows whose augmentation cancels the $(\mu_0 - \hat{\mu}_0)$ bias. The weighted estimator keeps the biased treated rows and discards their correction, collapsing onto the biased treated-only estimate — exactly the X-Learner of §3. The experiments in §6 confirm this: with the propensity known exactly, the inverse-variance-weighted

regression of the DR score has systematic bias 0.17 at $N_0=30$ (Table 2), indistinguishable from the X-Learner’s.

In summary: unweighted, the orthogonal score is unbiased but its variance becomes very large; weighted for efficiency, its variance is controlled but the bias is reintroduced. *For the DR score in the few-placebo regime, efficiency and unbiasedness cannot be achieved jointly*, and the common root of both consequences is the inverse-propensity factor $1/(1 - \pi)$ that the regime makes large.

Finally, we delimit the scope of this claim. It is *not* a refutation of Neyman-orthogonality or of the DR-Learner, whose asymptotic oracle properties — $\sqrt{N}$ -consistency and semiparametric efficiency — are not in dispute, and which were never advertised for extreme-allocation designs. Those oracle properties are statements about $N \rightarrow \infty$ *at fixed overlap*. The few-placebo regime violates that premise: it sends overlap to its edge *as* the regime is entered, so the asymptotics and the regime are coupled rather than separable. Our contribution is to map that boundary — a structural trade-off between asymptotic oracle properties, which assume fixed overlap, and finite-sample behaviour under structural regime asymmetry. Interpreted in this way, the result indicates the resolution: not a better orthogonal score, but an estimator that does not depend on overlap at all, with no inverse-propensity factors. We develop such an estimator in the next section.

5 Gaussian-Process Estimation of the Conditional Average Treatment Effect

The X-Learner (§3) and the doubly-robust score (§4) fail for what look like different reasons — a first-order bias in one case, an overlap-driven remainder and a weighting conflict in the other. They share one root. Each forms a *point estimate* $\hat{\mu}_0$ of the placebo nuisance — a regularized tree fit on the few control units — and then attaches a variance to the second-stage regression *around that point estimate*. But a point estimate carries a regularization bias, and that bias is invisible to any variance computed after the fact: the variance describes scatter about $\hat{\mu}_0$, not the distance from $\hat{\mu}_0$ to μ_0 . No second-stage noise model, orthogonal or not, can see it. The same defect afflicts any remedy that keeps a point estimate at its core — including one that estimates the regularization bias itself and subtracts it, since the bias estimate is then a point estimate in turn.

The remedy is not a better point estimate, nor a separate model of its bias. It is to dispense with the point estimate altogether. Concretely, GP-CATE models each arm’s outcome surface with a Gaussian process. For arm $w \in \{0, 1\}$, fit a GP to $\{(X_i, Y_i) : W_i = w\}$ with a kernel

$$k_w(x, x') = \sigma_w^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell_w^2}\right) + \eta_w^2 \mathbb{I}[x = x'], \quad (9)$$

an isotropic squared-exponential plus observation noise, with $(\sigma_w, \ell_w, \eta_w)$ set by marginal likelihood (empirical Bayes). The GP returns a full posterior over the *latent function* μ_w — a mean $m_w(x)$ and a pointwise standard deviation $s_w(x)$ (the observation-noise term removed, so s_w is uncertainty about the function, not about a future outcome). The CATE posterior is the difference of the two arm posteriors,

$$\tau(x) \mid \mathcal{D} \sim \mathcal{N}(m_1(x) - m_0(x), s_1(x)^2 + s_0(x)^2), \quad (10)$$

and the $(1 - \alpha)$ credible interval is $m_1(x) - m_0(x) \pm z_{\alpha/2} \sqrt{s_1(x)^2 + s_0(x)^2}$. This completes the specification of the method.

Before turning to its properties, we note that a Bayesian T-learner with GP arms is not itself a new estimator — it is close to the multi-task GP model of [Alaa and van der Schaar \(2017\)](#), without the cross-arm coupling. The contribution of this paper is not the estimator

Algorithm 1: GP-CATE.

Input: $(X_i, W_i, Y_i)_{i=1}^N$.

1. Fit a GP with kernel (9) to the treated units; set $(\sigma_1, \ell_1, \eta_1)$ by marginal likelihood.
2. Fit a GP likewise to the control units, giving $(\sigma_0, \ell_0, \eta_0)$.
3. At any x , read off the latent-function posteriors $m_w(x), s_w(x)$ for $w = 0, 1$.

Output: CATE posterior (10) and its credible interval.

but the preceding analysis: the diagnosis and the negative result identify *why* the meta-learner machinery mis-calibrates in the few-placebo regime, and GP-CATE is the estimator that the root-cause analysis singles out.

What makes this construction calibrated where the meta-learners are not? The GP posterior is calibrated *by construction* in a sense the post-hoc variances of §3–§4 are not. Its posterior mean m_0 is also shrunk — toward the prior, hence “biased” — but the GP posterior variance $s_0(x)^2$ *is the variance of that shrinkage*: posterior mean and posterior variance are two aspects of a single coherent object, so the uncertainty s_0 accounts for precisely the gap that a point estimate leaves unmodelled. Specifically, $s_0(x)$ is large where the control units are sparse and small where they are dense; in the few-placebo regime s_0 is large over much of the covariate space, and the CATE interval (10) is correspondingly — and correctly — wide. Under standard smoothness conditions GP posteriors contract at minimax rates and their credible sets attain frequentist coverage at the nominal level (Ghosal and van der Vaart, 2007; Rasmussen and Williams, 2006); the empirical-Bayes kernel adapts the prior to the data. GP-CATE needs no cross-fitting, no orthogonal score, and no bias model; its single modelling choice is the kernel family.

A word on computation. An exact GP costs $\mathcal{O}(N_w^3)$ per arm. For the control arm this is free: the few-placebo regime is *defined* by N_0 small, so the $\mathcal{O}(N_0^3)$ control fit is negligible. The treated arm is the concern — in the motivating applications (large-scale A/B tests, registry studies) N_1 can run to millions, and an exact treated GP is then infeasible. It is also unnecessary, as the structure of the problem indicates. The treated arm is, by construction, the well-determined one; its posterior contributes only a small share of the CATE variance $s_1(x)^2 + s_0(x)^2$. We measure that share at 6–25 % in the few-placebo regime (see the IHDP results in §6 and Appendix B) — the control term dominates. The treated posterior therefore need only be *approximately* right. We recommend, as the method’s default for large N_1 : fit the control arm with an exact GP, and the treated arm with any scalable surrogate — a sparse / inducing-point GP (Rasmussen and Williams, 2006), or a frequentist learner with a delta-method variance. §6 confirms that this is safe: capping the treated GP at 80 of 500 units leaves coverage at 0.97, unchanged from the full fit. The calibrated-uncertainty machinery is essential only where the data are scarce — the control arm — and there it is also inexpensive.

Two caveats about GP-CATE’s calibration are worth stating in advance of the experiments. First, when μ_w is smooth and d small, the empirical-Bayes GP is slightly cautious and coverage runs a little above nominal (0.97–0.99 in §6); this is the safe direction. Second, with few controls in a high-dimensional covariate space, $s_0(x)$ approaches the prior scale over most of the space, because the data genuinely under-determines μ_0 there; the credible interval is then wide. We regard this as a property of the method, not a deficiency: it is the correct statement that thirty controls in twenty-five dimensions cannot determine the placebo surface, and §6 shows that the alternatives’ narrower intervals in that setting do not cover. GP-CATE reports the difficulty of the inference problem rather than concealing it.

6 Experiments

The experiments follow the paper’s three claims, in order: the X-Learner under-covers and the cause is bias (Table 1); the orthogonal score does not repair it (Table 2); and GP-CATE is calibrated where the standard interval estimators are not, across three designs (Table 3, Figure 1, Table 4).

Coverage requires a known $\tau(x)$, so we evaluate on three designs. The *linear* design has $X \sim \mathcal{N}(0, I_4)$, $W \mid X \sim \text{Bernoulli}(\text{logistic}(0.4X_0))$, $Y = 0.5X_0 + 0.3X_1 + \mathcal{N}(0, 0.5^2) + W\tau(X)$ with the linear $\tau = 1 + 0.3X_0 - 0.2X_2$. The *non-linear* design has a two-covariate propensity, an interaction-laden response surface, and the non-linear $\tau = 1 + \sin(1.4X_0) - 0.3X_2^2$. Finally, *IHDP* is the semi-synthetic benchmark of Hill (2011) — 25 real covariates and a simulated response surface with known per-unit τ — which we put into the few-placebo regime by keeping all treated units and sub-sampling the control arm. The synthetic designs fix $N_1 = 500$ and sweep N_0 ; IHDP sweeps the sub-sampled N_0 . Full details are in Appendix B.

We compare three families of estimators. The diagnosis and the negative result concern the standard X-Learner (Künzel et al., 2019) and the doubly-robust score / DR-Learner (Kennedy, 2023); for these we measure the systematic bias each pseudo-outcome carries. The calibration comparison is against the two most common CATE estimators that provide uncertainty intervals: *Causal Forest* — the causal-forest DML estimator with honest confidence intervals (Athey and Imbens, 2016; Wager and Athey, 2018) — and *BART*, an S-learner Bayesian additive regression tree whose CATE posterior is that of $f(x, 1) - f(x, 0)$ (Hill, 2011). For point accuracy we additionally report the S-, T-, X-, and DR-Learner. No post-hoc calibration is applied to any method.

With the designs and estimators in place, we now report the results in the order of the paper’s three claims. We turn first to the diagnosis. Table 1 decomposes, over seeds, the error of a Bayesian regression on the X-Learner pseudo-outcome — the natural way to equip the X-Learner with an interval (§3) — into a systematic and a sampling component. The systematic bias is a large fraction of the posterior spread at small N_0 and decays as the placebo arm grows, the behaviour (5) predicts. A posterior displaced by 0.55 standard deviations cannot cover: the nominal-95 % interval covers 0.88 at $N_0=30$, recovering to 0.97 by $N_0=500$ as the bias decays.

Table 1: Bias/spread decomposition of a Bayesian regression on the X-Learner pseudo-outcome (linear design), with the resulting credible-interval coverage. The systematic bias is a large fraction of the posterior spread at $N_0=30$ and the interval under-covers.

	bias (RMSE)	spread	coverage
$N_0 = 30$	0.146	0.268	0.88
$N_0 = 500$	0.028	0.108	0.97

Next, Table 2 measures the negative result of §4. The doubly-robust score’s systematic bias is essentially identical to the X-Learner pseudo-outcome’s (0.154 vs 0.146 at $N_0=30$): orthogonalization yields no improvement here. The bias persists when the propensity is *known exactly* (0.174 at $N_0=30$) — the efficiency/unbiasedness conflict of §4, not propensity-estimation error — and recedes only as the design balances, i.e. as overlap is restored. The unweighted DR-Learner avoids the bias, but its variance is very large: $\sqrt{\varepsilon_{\text{PEHE}}} = 0.63$ at $N_0=30$, the largest in Table 5.

We now turn to the main calibration result. Table 3 and Figure 1 compare GP-CATE against the two standard interval estimators. GP-CATE is the only estimator we test that is calibrated throughout: coverage 0.94–0.99 on every design and sample size. The two most common interval-producing CATE estimators perform poorly in this regime. *Causal Forest*’s honest confidence intervals under-cover substantially — 0.59–0.83 on the synthetic designs and 0.34–0.45 on IHDP. *BART* is calibrated on the linear design (0.91–0.94) but under-covers on

Table 2: Systematic bias of an inverse-variance-weighted regression of the doubly-robust score (linear design), propensity estimated and known exactly. Neither improves on the X-Learner pseudo-outcome’s 0.146 at $N_0=30$ (Table 1).

N_0	30	100	500
DR score, estimated propensity	0.154	0.099	0.049
DR score, propensity known exactly	0.174	0.086	0.033

Table 3: Coverage of nominal-95 % credible intervals, no post-hoc calibration. Columns are small/mid/large N_0 per design (linear and non-linear: 30/100/500; IHDP: 30/60/100). GP-CATE is calibrated throughout; both standard interval estimators under-cover on the harder designs.

method	linear			non-linear			IHDP		
	s	m	l	s	m	l	s	m	l
Causal Forest (Wager and Athey, 2018)	0.79	0.76	0.83	0.59	0.61	0.63	0.45	0.40	0.34
BART (Hill, 2011)	0.94	0.94	0.91	0.79	0.86	0.82	0.59	0.74	0.77
GP-CATE (ours)	0.97	0.99	0.98	0.94	0.94	0.95	0.96	0.96	0.95

the non-linear (0.79–0.86) and IHDP (0.59–0.77) designs. GP-CATE is mildly conservative on the smooth linear design (0.97–0.99) — the safe direction, as §5 anticipates — and essentially nominal on the non-linear and IHDP designs.

Among the three designs, IHDP warrants the closest examination: with 25 real covariates and few controls, it is the most demanding test. Table 4 pairs coverage with interval width. GP-CATE is the only calibrated estimator (0.95–0.96), and its intervals are wide (≈ 12 –18). This width is the accurate report: with thirty controls in twenty-five dimensions the placebo surface is near-undetermined, and the GP posterior reflects this. It is *not* a vague prior reasserting itself — the estimator is performing valid inference, not abdicating. The control-arm GP posterior variance contracts to 0.55 of its (proper, fitted) prior at $N_0=30$ and to 0.16 at $N_0=100$; on the lower-dimensional linear and non-linear designs the contraction reaches 0.05–0.07 (Appendix B). The GP extracts useful information from the few controls, but in twenty-five dimensions cannot extract enough to make the interval narrow. Causal Forest and BART report intervals a third to a quarter as wide and under-cover substantially — Causal Forest’s 3–4-wide intervals cover only 0.34–0.45. Their intervals are narrow precisely because they are miscalibrated. A practitioner is better served by GP-CATE’s wide, calibrated interval, which indicates that more control data or a lower-dimensional analysis is required, than by a narrow interval that miscovers a third of the time.

Beyond coverage and width, we also assess point accuracy: calibration is not achieved at the expense of estimation error. Table 5 reports $\sqrt{\varepsilon_{\text{PEHE}}}$ on the linear design: GP-CATE is the most accurate estimator at $N_0 \geq 100$ and competitive at $N_0=30$, where BART edges it. On the non-linear design GP-CATE’s $\sqrt{\varepsilon_{\text{PEHE}}}$ is 0.51/0.33/0.25 across the sweep, on a par with BART (0.51/0.38/0.28) and ahead of Causal Forest (0.65/0.49/0.39). On IHDP, BART is in fact the most accurate *point* estimator ($\sqrt{\varepsilon_{\text{PEHE}}} \approx 2.4$ –3.7 versus GP-CATE’s 3.5–4.5) — but its intervals under-cover (Table 4). In summary, GP-CATE is competitive on accuracy across all designs and is the only method that is also calibrated; where a standard estimator is more accurate, it is so while reporting intervals that do not cover.

Taken together, these experiments establish the paper’s three claims. The X-Learner under-covers in the few-placebo regime from first-order nuisance bias (Table 1); the orthogonal-score fix does not remove that bias, even with the propensity known (Table 2). GP-CATE, which assigns the hard-to-learn nuisance a full Bayesian posterior rather than a point estimate, is

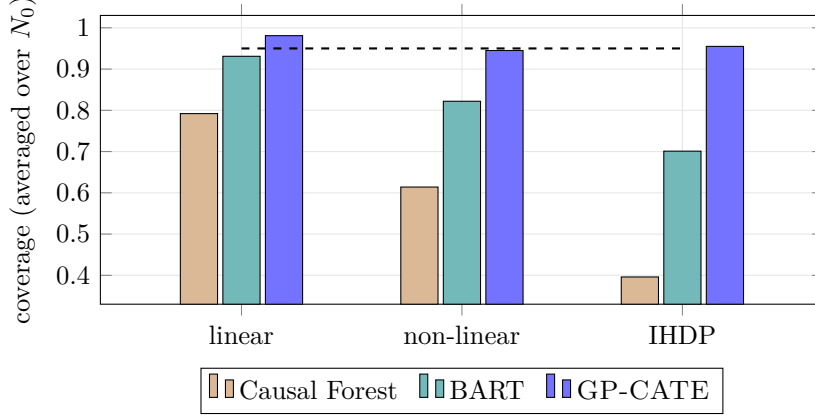


Figure 1: Coverage of nominal-95 % intervals, averaged over the N_0 sweep, on the three designs. GP-CATE (blue) stays at or above the nominal line (dashed); the two standard interval estimators — Causal Forest and BART — under-cover, most on IHDP.

Table 4: IHDP: coverage and mean interval width, N_0 sub-sampled. GP-CATE covers; both standard interval estimators under-cover, at a third to a quarter of the width.

N_0	coverage (target 0.95)			mean width		
	30	60	100	30	60	100
Causal Forest	0.45	0.40	0.34	4.0	3.5	3.0
BART	0.59	0.74	0.77	3.8	3.8	3.7
GP-CATE (ours)	0.96	0.96	0.95	15.5	17.9	12.2

calibrated across the linear, non-linear, and IHDP designs (Table 3) and accurate (Table 5) — the only calibrated method we test, the standard Causal Forest and BART included. Its cost is interval width on high-dimensional covariates (Table 4), reflecting the true difficulty of the problem rather than concealing it.

7 Discussion

In the few-placebo regime the practical choice of estimator is governed by calibration, and GP-CATE is the only estimator we tested that is calibrated across all three designs (§6). The standard interval-producing estimators each fail somewhere: Causal Forest under-covers severely, down to 0.34 on IHDP — a misplaced interval, the dangerous direction — and BART, calibrated on the smooth linear design, under-covers once the response surface is non-linear or high-dimensional. We therefore recommend GP-CATE as the default when one arm is genuinely small; once the arms approach balance the few-placebo problem dissolves and a plain X-Learner is adequate and cheaper.

However, GP-CATE’s intervals on the 25-dimensional IHDP covariates are wide — several times the spread of τ itself. We have not treated this as a defect to be engineered away. Thirty control units in twenty-five dimensions do not determine the placebo surface, and a calibrated interval must reflect this; the alternatives’ narrower IHDP intervals are narrower precisely because they under-cover (§6). A practitioner is better served by a wide, calibrated interval than by a narrow one that miscovers: the wide interval correctly indicates that more control data, or a lower-dimensional analysis, are required before a confident claim. The width is, however, a real limitation on what can be concluded from such data, and reducing it without sacrificing coverage — by borrowing strength across arms with a coupled GP (Alaa and van der Schaar,

Table 5: $\sqrt{\varepsilon_{\text{PEHE}}}$ on the linear design, $N_1 = 500$ (lower is better). GP-CATE is the most accurate at $N_0 \geq 100$.

N_0	30	100	500
S-Learner	0.36	0.24	0.18
T-Learner	0.46	0.37	0.27
X-Learner	0.58	0.23	0.11
DR-Learner (unweighted)	0.63	0.49	0.18
Causal Forest	0.31	0.24	0.16
BART	0.23	0.20	0.15
GP-CATE (ours)	0.29	0.13	0.07

2017), or by a data-driven dimension reduction — is the most useful direction for future work.

More generally, the diagnosis and the negative result share a principle that extends beyond the few-placebo regime: a variance attached to a point estimate of a hard-to-learn nuisance cannot be expected to yield calibration. A point estimate’s regularization bias is invisible to any variance computed around it, whether that variance comes from a homoskedastic likelihood, an errors-in-variables weighting, an orthogonal score, or an explicit bias model. The reliable route to a calibrated downstream interval is a nuisance that carries its own posterior. GP-CATE is one instantiation; the principle is broader.

Several strands of related work bear on this picture. The X-Learner is [Künzel et al. \(2019\)](#); the doubly-robust score and DR-Learner are [Robins et al. \(1995\)](#) and [Kennedy \(2023\)](#); cross-fitting and the orthogonality framework are [Chernozhukov et al. \(2018\)](#). That orthogonal estimators degrade under poor overlap is known for the average effect; our contribution is to show the few-placebo regime *is* a poor-overlap regime by construction, and to add the propensity-independent failure of §4. Gaussian-process and Bayesian-nonparametric models for treatment effects are established — the multi-task GP of [Alaa and van der Schaar \(2017\)](#), and BART-based BCF ([Hahn et al., 2020](#)); GP-CATE is a deliberately plain instance, and our point is diagnostic rather than architectural: a fully-Bayesian nuisance is what the few-placebo regime requires. We benchmark directly against the two most common interval-producing CATE estimators — Causal Forest ([Athey and Imbens, 2016](#); [Wager and Athey, 2018](#)) and BART ([Hill, 2011](#)) — and find both under-cover in this regime (§6): Causal Forest’s honest asymptotic intervals and BART’s posterior are calibrated under good overlap but not when one arm is small. Generalized-Bayes and bootstrap calibrations ([Javurek et al., 2026](#); [Lin and Han, 2026](#); [Lyddon et al., 2019](#)) and semiparametric Bernstein–von Mises results ([Breunig et al., 2025](#); [Ray and van der Vaart, 2020](#)) target settings with good overlap; conformal meta-learners ([Alaa et al., 2023](#); [Lei and Candès, 2021](#)) give finite-sample intervals for the individual effect rather than a posterior for the CATE function.

Four limitations of GP-CATE merit explicit mention. First, an exact GP costs $\mathcal{O}(N_w^3)$ per arm. The control arm, small by definition, is inexpensive; for a large treated arm §5 provides a concrete prescription — a sparse surrogate, justified because the treated arm carries only 6–25 % of the CATE variance — so this is a resolved engineering choice, not an obstacle to practical use. Second, GP-CATE is mildly conservative on smooth, low-dimensional designs (§6); the safe direction, but a mild efficiency loss. Third, in high dimension with very few controls the calibrated interval can be wide enough to limit what the data support — a property of the problem, made explicit rather than concealed. Fourth, the kernel family is a modelling choice; we use an isotropic squared-exponential, since per-dimension length scales over-fit from a small control arm (§6).

8 Conclusion

In the few-placebo regime, the X-Learner with a Bayesian second stage produces credible intervals that under-cover. The cause is structural: the X-Learner pseudo-outcome inherits the bias of the nuisance model fitted to the small control arm, and a posterior centered on a biased target is displaced rather than merely widened. The standard correction — an orthogonal doubly-robust score — does not resolve this, because the few-placebo regime is itself a poor-overlap regime: the score’s nominally second-order remainder is amplified there, and the efficient inverse-variance-weighted variant trades that amplification back for the bias. Bias and variance cannot be controlled jointly through orthogonalization alone.

Both failures share a single cause. Each attaches a variance to a *point estimate* of the placebo nuisance, and a point estimate’s regularization bias is invisible to any variance computed around it. GP-CATE replaces the point estimate with a per-arm Gaussian process, so the small arm’s uncertainty enters the CATE posterior directly: the posterior mean and the posterior variance are two aspects of a single coherent object, rather than a point estimate with a separately attached error bar. The CATE posterior is the difference of the two arm posteriors, and the construction requires no cross-fitting, no orthogonal score, and no auxiliary bias model.

Across a linear design, a non-linear design, and the IHDP semi-synthetic benchmark, GP-CATE is the only estimator we evaluate that attains calibrated coverage throughout; Causal Forest, BART, and the standard meta-learners do not. The cost is interval width: when few controls do not determine the placebo surface in high dimension, GP-CATE’s intervals widen accordingly — the accurate answer that the data support, in contrast to the narrower but miscalibrated intervals of the alternatives.

References

- A. M. Alaa and M. van der Schaar. Bayesian inference of individualized treatment effects using multi-task gaussian processes. In *Advances in Neural Information Processing Systems*, 2017.
- A. M. Alaa, Z. Ahmad, and M. van der Laan. Conformal meta-learners for predictive inference of individual treatment effects. In *Advances in Neural Information Processing Systems*, 2023.
- S. Athey and G. Imbens. Recursive partitioning for heterogeneous causal effects. *Proceedings of the National Academy of Sciences*, 113(27):7353–7360, 2016.
- C. Breunig, R. Liu, and Z. Yu. Double robust Bayesian inference on average treatment effects. *Econometrica*, 93(2):539–568, 2025.
- V. Chernozhukov, D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.
- S. Ghosal and A. van der Vaart. Convergence rates of posterior distributions for noniid observations. *The Annals of Statistics*, 35(1):192–223, 2007.
- P. R. Hahn, J. S. Murray, and C. M. Carvalho. Bayesian regression tree models for causal inference: Regularization, confounding, and heterogeneous effects (with discussion). *Bayesian Analysis*, 15(3):965–1056, 2020.
- J. L. Hill. Bayesian nonparametric modeling for causal inference. *Journal of Computational and Graphical Statistics*, 20(1):217–240, 2011.
- E. Javurek, D. Frauen, Y. Wang, and S. Feuerriegel. Generalized Bayes for causal inference. *arXiv preprint arXiv:2603.03035*, 2026.

- E. H. Kennedy. Towards optimal doubly robust estimation of heterogeneous causal effects. *Electronic Journal of Statistics*, 17(2):3008–3049, 2023.
- S. R. Künzel, J. S. Sekhon, P. J. Bickel, and B. Yu. Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences*, 116(10):4156–4165, 2019.
- L. Lei and E. J. Candès. Conformal inference of counterfactuals and individual treatment effects. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 83(5):911–938, 2021.
- Z. Lin and F. Han. Bootstrap consistency for general double/debiased machine learning estimators. *arXiv preprint arXiv:2604.17239*, 2026.
- S. P. Lyddon, C. C. Holmes, and S. G. Walker. General Bayesian updating and the loss-likelihood bootstrap. *Biometrika*, 106(2):465–478, 2019.
- C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.
- K. Ray and A. van der Vaart. Semiparametric Bayesian causal inference. *The Annals of Statistics*, 48(5):2999–3020, 2020.
- J. M. Robins, A. Rotnitzky, and L. P. Zhao. Semiparametric efficiency in multivariate regression models with missing data. *Journal of the American Statistical Association*, 90(429):106–121, 1995.
- P. R. Rosenbaum and D. B. Rubin. The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1):41–55, 1983.
- S. Wager and S. Athey. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242, 2018.

A Derivation of the Doubly-Robust Remainder

We derive equation (6). Fix the cross-fitted nuisances $\hat{\mu}_0, \hat{\mu}_1, \hat{\pi}$ and take expectations over (W, Y) given X . With $\mathbb{E}[W | X] = \pi$ and $\mathbb{E}[Y | X, W=w] = \mu_w$,

$$\begin{aligned}\mathbb{E}\left[\frac{W(Y-\hat{\mu}_1)}{\hat{\pi}} \mid X\right] &= \frac{\pi}{\hat{\pi}}(\mu_1 - \hat{\mu}_1), \\ \mathbb{E}\left[\frac{(1-W)(Y-\hat{\mu}_0)}{1-\hat{\pi}} \mid X\right] &= \frac{1-\pi}{1-\hat{\pi}}(\mu_0 - \hat{\mu}_0).\end{aligned}$$

Substituting into (3) and using $\hat{\mu}_w = \mu_w - (\mu_w - \hat{\mu}_w)$,

$$\begin{aligned}\mathbb{E}[D^{\text{DR}} | X] &= (\mu_1 - \mu_0) + \left(\frac{\pi}{\hat{\pi}} - 1\right)(\mu_1 - \hat{\mu}_1) \\ &\quad - \left(\frac{1-\pi}{1-\hat{\pi}} - 1\right)(\mu_0 - \hat{\mu}_0).\end{aligned}$$

The first term is $\tau(X)$; the rest is $R(X)$ of (6), each summand a product of a propensity error and an outcome-nuisance error — the second-order property. The arm-conditional means (8) follow by not averaging over W . The first-order expansion $\frac{1-\pi}{1-\hat{\pi}} - 1 = \frac{\hat{\pi}-\pi}{1-\hat{\pi}} \approx \frac{\hat{\pi}-\pi}{1-\pi}$ holds for $|\hat{\pi} - \pi| \ll 1 - \pi$ and gives (7); the $1/(1 - \pi)$ factor is the overlap amplification.

B Implementation and Reproducibility

The three designs are specified as follows. The *linear* design uses $X \sim \mathcal{N}(0, I_4)$, $W \mid X \sim \text{Bernoulli}(\text{logistic}(0.4X_0))$, $Y = 0.5X_0 + 0.3X_1 + 0.5\xi + W\tau(X)$ with $\tau = 1 + 0.3X_0 - 0.2X_2$. The *non-linear* design has a two-covariate propensity, an interaction-laden response surface, and a non-linear $\tau = 1 + \sin(1.4X_0) - 0.3X_2^2$. *IHDP* is the semi-synthetic benchmark of Hill (2011) — 25 real covariates, 747 units, and a simulated response surface with known per-unit τ ; we keep all treated units and sub-sample the control arm to N_0 to mirror $N_0 \ll N_1$. The few-placebo regime in the synthetic designs is produced by stratified sampling to N_0 controls and $N_1=500$ treated; the known-propensity check of §6 draws $W \sim \text{Bernoulli}(p)$ with p fixed, so $\pi \equiv p$.

For the estimators themselves, GP-CATE fits, per arm, a Gaussian process with a constant-times-RBF-plus-white-noise kernel, hyperparameters by marginal likelihood with two restarts; the RBF length scale is isotropic (per-dimension ARD length scales over-fit a small control arm and *worsen* coverage — on IHDP, to 0.78). The bias measurements of §6 regress the X-Learner pseudo-outcome and the doubly-robust score, with gradient-boosted nuisances (depth 3, 200 estimators) and $K=5$ cross-fitting. The interval baselines are the econml causal-forest DML estimator (500 trees, gradient-boosted nuisance models, honest confidence intervals via `effect_interval`) and an S-learner BART on $[X, W]$ (60 trees, 200 posterior draws after 100 burn-in), its CATE interval taken from the posterior of $f(x, 1) - f(x, 0)$; the S-, T-, X-, and DR-Learner use the same gradient-boosted nuisances.

Synthetic-design numbers average 20 seeds: coverage, $\sqrt{\varepsilon_{\text{PEHE}}}$, and width on 1,500 fresh test points, and systematic bias as the RMSE of the seed-averaged prediction against τ on a fixed 3,000-point set. IHDP averages the 10 response-surface replications, each with 4 control sub-samples, with coverage evaluated against the known per-unit τ at all 747 covariate points.

The posterior-contraction ratios reported in §6 and the treated-arm variance share discussed in §5 compare, per arm, the fitted prior function variance — the `ConstantKernel` amplitude σ_w^2 — with the mean posterior function variance $s_w(x)^2$ over the test set; the contraction ratio is their quotient. The contraction is 0.05–0.07 on the linear and non-linear designs and 0.55 ($N_0=30$) to 0.16 ($N_0=100$) on IHDP; the treated arm’s share $s_1^2/(s_0^2 + s_1^2)$ is 0.06–0.25 in the few-placebo regime. As a sanity check, we also fit the treated GP on a random subset of the treated arm; coverage on the linear design at $N_0=30$ is 0.97 for caps of 500, 200, and 80.